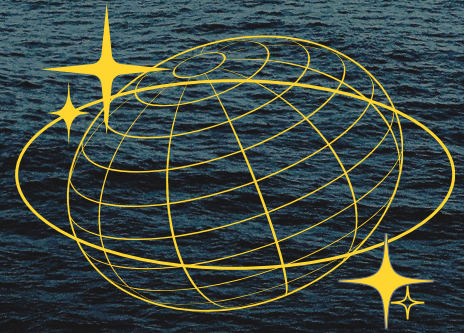


Cybersecurity Reconnaissance & Network Discovery Report

Attack Surface Security Assessment



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Program	Cybersecurity Professional Program
Week	Week 02
Project	Network Scanning & Footprinting

1. Executive Summary & Assessment Overview

This report details the findings of a cybersecurity reconnaissance and attack surface security assessment. The scope of this assessment included Open Source Intelligence (OSINT) gathering, DNS and WHOIS lookups, web application fingerprinting, network scanning, and subnet discovery for the targets: networkwalks.com, microsoft.com, 192.168.56.0/24, and 10.153.99.0/24. The objective was to identify potential vulnerabilities and provide actionable recommendations for enhancing the security posture of these networks and their associated assets.

2. External Target Footprinting: networkwalks.com

Domain Registration & DNS Diagnostics

- **Registrar:** GoDaddy
- **Creation Date:** 2019-11-06T22:51:46Z
- **Expiry Date:** 2027-11-06T22:51:46Z
- **Privacy Guard:** Enabled
- **Name Servers:**
NS6135.HOSTGATOR.COM,
NS6136.HOSTGATOR.COM
- **DNS Resolution:** Resolved via 8.8.8.8 to IP address 192.232.216.135.

Web Infrastructure & Technology Stack

- **Web Server:** Apache HTTP Server
- **CMS:** WordPress 7.1
- **Plugins:** WordPress Download Manager plugin v3.3.58
- **Frontend Framework:** Bootstrap 7.1
- **JavaScript Library:** jQuery 3.7.1
- **Web Application Firewall (WAF):** ModSecurity detected via wafw00f.
- **Session Cookies:** Identified, indicating active user sessions.

3. External Attack Surface Analytics: microsoft.com

OSINT Enumeration & Security Contacts

OSINT enumeration using theHarvester revealed several Autonomous System Numbers (ASNs) associated with Microsoft:

- AS13335
- AS8075
- AS8070
- AS133618
- AS206834
- AS40034

Key security contact emails identified include:

- secure@microsoft.com
- opencode@microsoft.com
- dotnet-docker-bot@microsoft.com

IP Addresses & Subdomain Categorization

The analysis identified **123 external IP addresses** and **9,966 subdomains**. These were broadly categorized into:

- **Identity & Access Management:** Services related to user authentication and authorization.
- **Internal/Pre-Production Environments:** Subdomains potentially associated with development, testing, or staging environments.
- **AI/Cognitive API Services:** Endpoints for artificial intelligence and cognitive computing functionalities.

4. Internal Network Discovery & Node Telemetry

Local Interface Configuration

- **IPv4 Address:** 192.168.56.1
- **Subnet Mask:** 255.255.255.0

Zenmap/Nmap Ping Sweeps

- **192.168.56.0/24 Network:** 1 host reported as up.
- **10.153.99.0/24 Network:** 2 hosts reported as up:
 - 10.153.99.24
 - 10.153.99.99

Network Topology Mapping

(Detailed network topology map would be inserted here if visual representation was possible within this format. The discovered nodes and their interconnections form the basis of the network topology.)

5. Security Posture Evaluation & Hardening Recommendations

Based on the reconnaissance and discovery findings, the following recommendations are proposed to enhance the security posture:

- **Web Server/CMS Version Obscurity:** Suppress the disclosure of web server and Content Management System (CMS) version headers. This can help to reduce the attack surface by making it harder for attackers to identify known vulnerabilities associated with specific versions.
- **Private DNS Zones for Non-Production:** Implement private DNS zones for internal and non-production environments. This limits the visibility of these resources to authorized internal networks, reducing the risk of external exposure.
- **Subnet Isolation:** Enforce strict subnet isolation. Utilize firewalls and network access control lists (ACLs) to segment networks and control traffic flow between subnets, preventing lateral movement in case of a breach.
- **Npcap Driver Updates:** Ensure that the Npcap driver (or equivalent network packet capture driver) is kept up-to-date. Outdated drivers can sometimes lead to system instability or security vulnerabilities that could be exploited during comprehensive vulnerability scanning.

Conclusion

This report has provided a snapshot of the identified attack surface and network landscape for the assessed targets. Proactive implementation of the recommended security hardening measures is crucial for mitigating potential risks and strengthening the overall cybersecurity defenses.

Screenshots

Figure 1: Target Footprinting, DNS Resolution & Web Application Fingerprinting (networkwalks.com)

```
(kavi@kali)-[~]
$ curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
set-cookie: __wpdm_client=29ed9a0988ba79a834cceb9ea56f470f; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Sun, 23 Aug 2026 14:52:24 GMT
server: Apache
```

```
(kavi@kali)-[~]
$ nslookup networkwalks.com
;; communications error to 10.0.0.1#53: connection refused
;; communications error to 10.0.0.1#53: connection refused
;; communications error to 10.0.0.1#53: connection refused
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
;; communications error to 10.0.0.1#53: connection refused
;; communications error to 10.0.0.1#53: connection refused
;; communications error to 10.0.0.1#53: connection refused
```

```
(kavi@kali)-[~]  
$ wafw00f networkwalks.com
```



~ WAFW00F : v2.3.2 ~

The Web Application Firewall Fingerprinting Toolkit

```
[*] Checking https://networkwalks.com  
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.  
[~] Number of requests: 2
```

```
kavi@kali: ~  
Session Actions Edit View Help  
ding  
mining this data for your own personal or commercial purposes. Failure to comply with these terms  
may result in termination of access to the Whois database. These terms may be subject to modification  
at any time without notice.  
  
**NOTICE** This WHOIS server is being retired. Please use our RDAP service instead.  
  
(kavi@kali)-[~]  
$ whatweb networkwalks.com  
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpmc_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[__wpmc_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], UncommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]  
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1], Cookies[__wpmc_client], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpmc_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&pvtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]  
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1], Country[UNITED STATES][US], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&pvtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1]
```



```
kavi@kali: ~  
Session Actions Edit View Help  
  
(kavi@kali)-[~]  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2026-08-23T14:02:12Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/epp  
  
NOTICE: The expiration date displayed in this record is the date the  
registrar's sponsorship of the domain name registration in the registry is  
currently set to expire. This date does not necessarily reflect the expiration  
date of the domain name registrant's agreement with the sponsoring  
registrar. Users may consult the sponsoring registrar's Whois database to
```

Toolchain Used: whois, whatweb, nslookup, curl -I, wafw00f

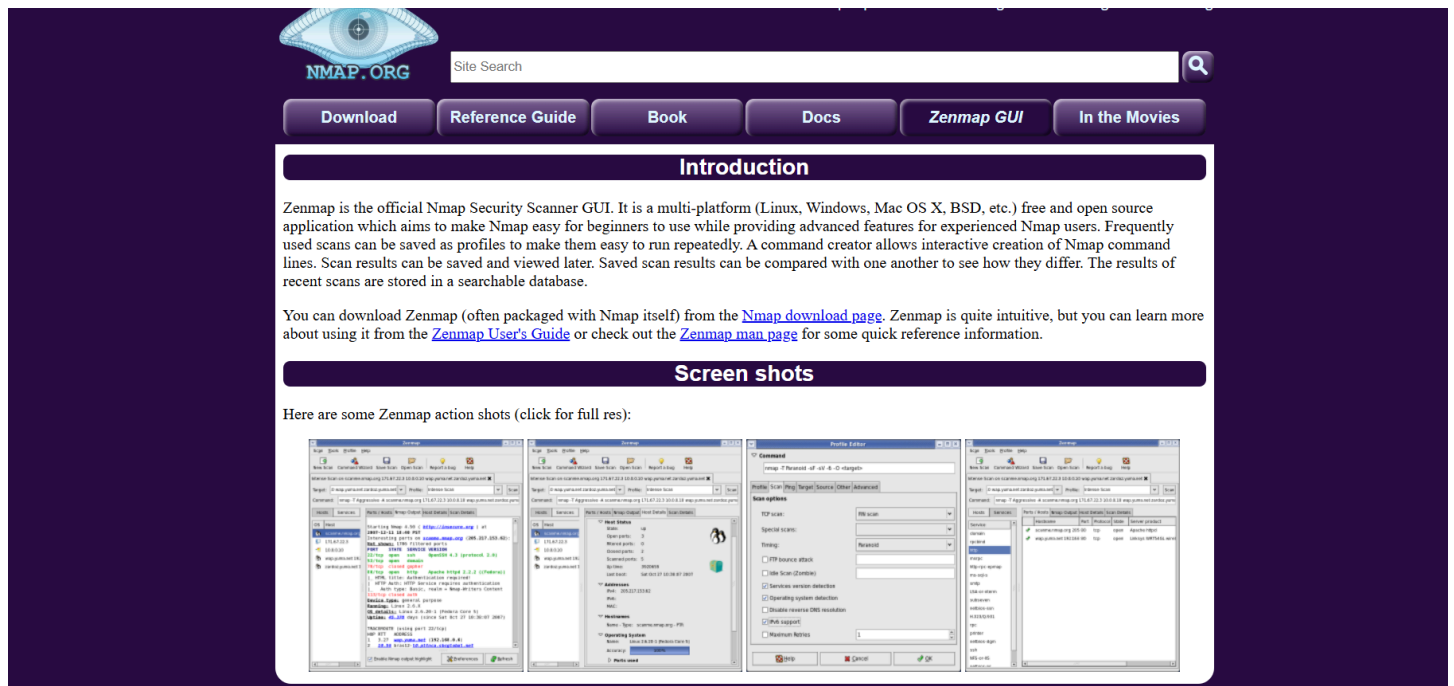
Technical Description & Observations:

- **WHOIS Record Analysis:** Demonstrates domain lifecycle verification via GoDaddy (Registered: Nov 6, 2019; Expires: Nov 6, 2027) with registrant identity obscured via Domains By Proxy, LLC.
- **DNS Enumeration:** Displays domain resolution to HostGator host IP 192.232.216.135 via public DNS resolver 8.8.8.8 after internal timeout on 10.0.0.1#53.
- **Web Stack Fingerprinting:** whatweb identifies WordPress 7.1 CMS, WordPress Download Manager plugin v3.3.58, Bootstrap 7.1, and jQuery 3.7.1.
- **HTTP Headers & WAF Diagnostics:** curl -I confirms HTTP/2 redirect rules, server tokens exposing Apache, and active session cookies (__wpdm_client). wafw00f identifies an active ModSecurity (SpiderLabs) Web Application Firewall guarding the perimeter.

Figure 2: Automated OSINT Asset Surface Enumeration (microsoft.com)

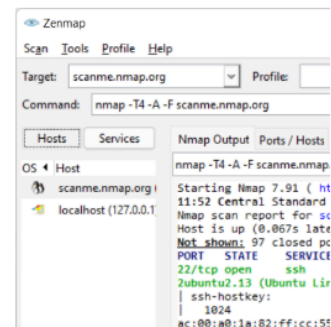
- Demonstrates a host-only virtual adapter environment lacking a default gateway, defining the local scope for internal broadcast sweeps.

Figure 4: Security Scanner Acquisition & Documentation Reference (Nmap / Zenmap)



[Copyright page.](#)

Microsoft Windows binaries



Please read the [Windows section](#) of the Install Guide for limitations and installation instructions for the Windows version of Nmap. It's provided as an executable self-installer which includes Nmap's dependencies and the Zenmap GUI. We support Nmap on Windows 7 and newer, as well as Windows Server 2008 R2 and newer. We also maintain a [guide for users who must run Nmap on earlier Windows releases](#).

Latest stable release self-installer: [nmap-7.991-setup.exe](#)

We have written [post-install usage instructions](#). Please [notify us](#) if you encounter any problems or have suggestions for the installer.

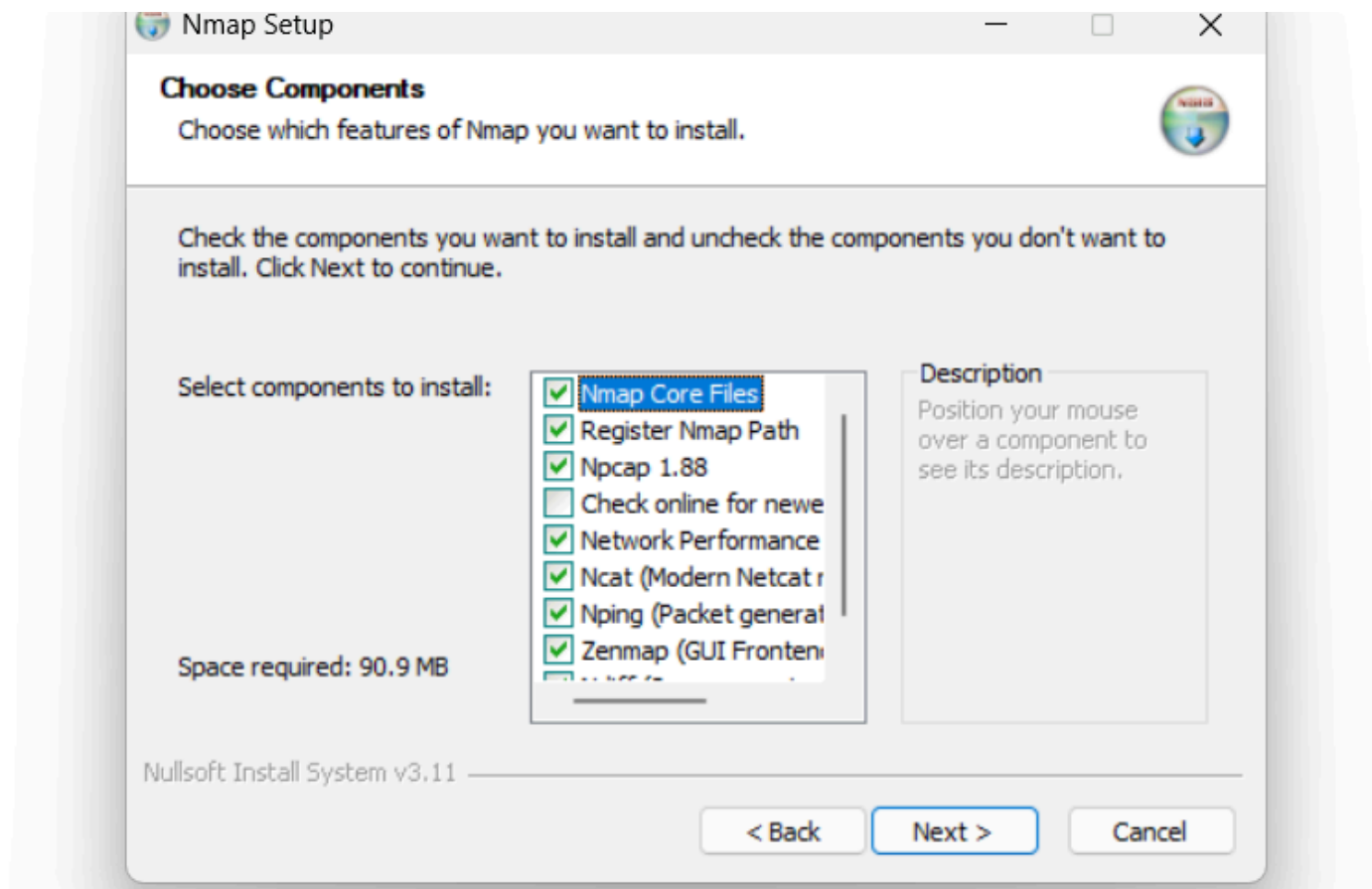
Linux RPM Source and Binaries

Toolchain / Environment: Nmap.org Official Download Portal & Reference Documentation

Technical Description & Observations:

- Captures the official distribution source for the Zenmap GUI frontend and Nmap suite installer (nmap-7.991-setup.exe) for Microsoft Windows platforms.
- References GUI features including command profile configuration, real-time scan output visualization, and database-backed scan comparison modules.

Figure 5: Zenmap Deployment & Component Dependency Selection

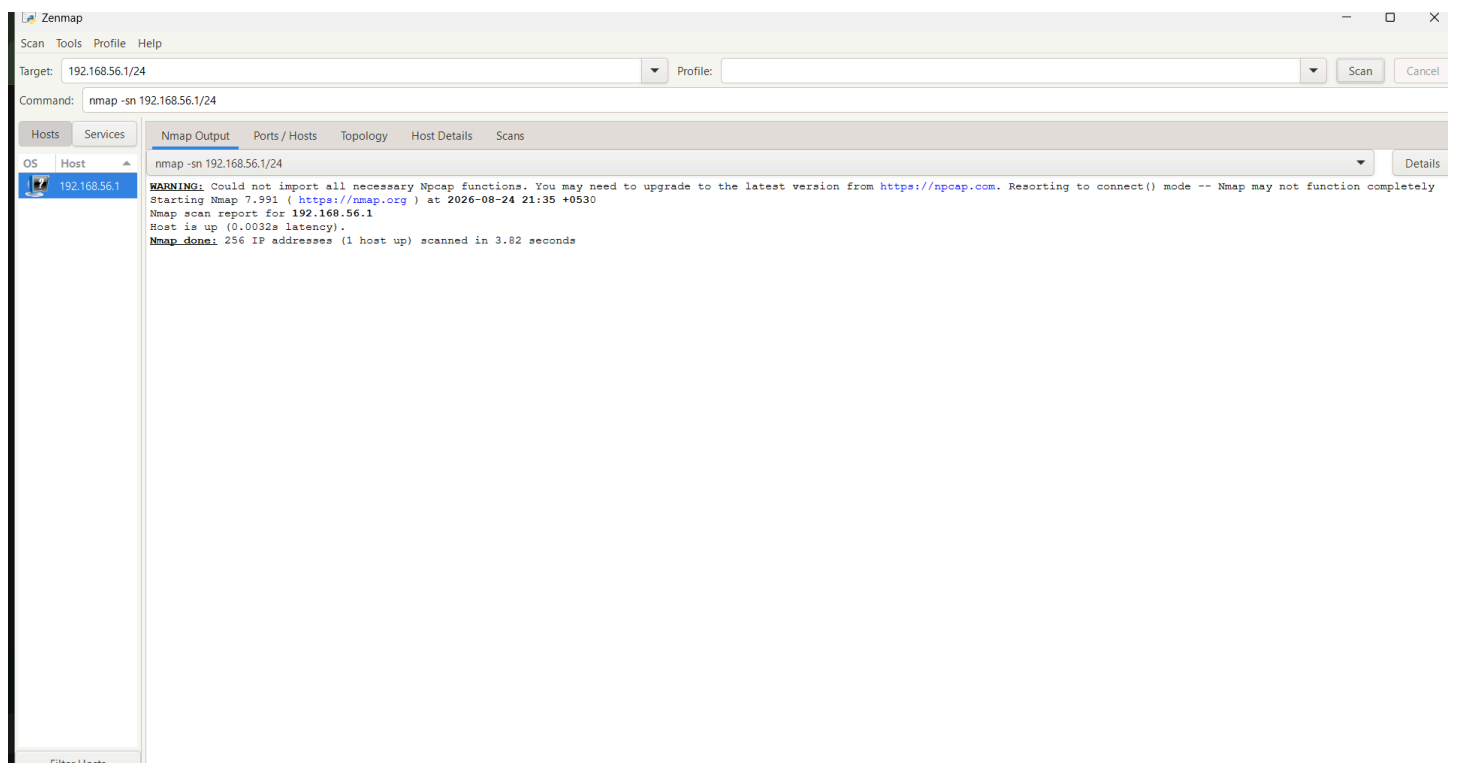


Toolchain / Environment: Nullsoft Install System (v3.11)

Technical Description & Observations:

- Illustrates the modular software configuration setup for the Nmap toolset.
- Highlights deployment selection for core scanner binaries, system path environment registration, Npcap packet capture driver integration (v1.88), Netcat modern (ncat), and the Zenmap GUI interface.

Figure 6: Host-Only Subnet Ping Sweep (192.168.56.0/24)

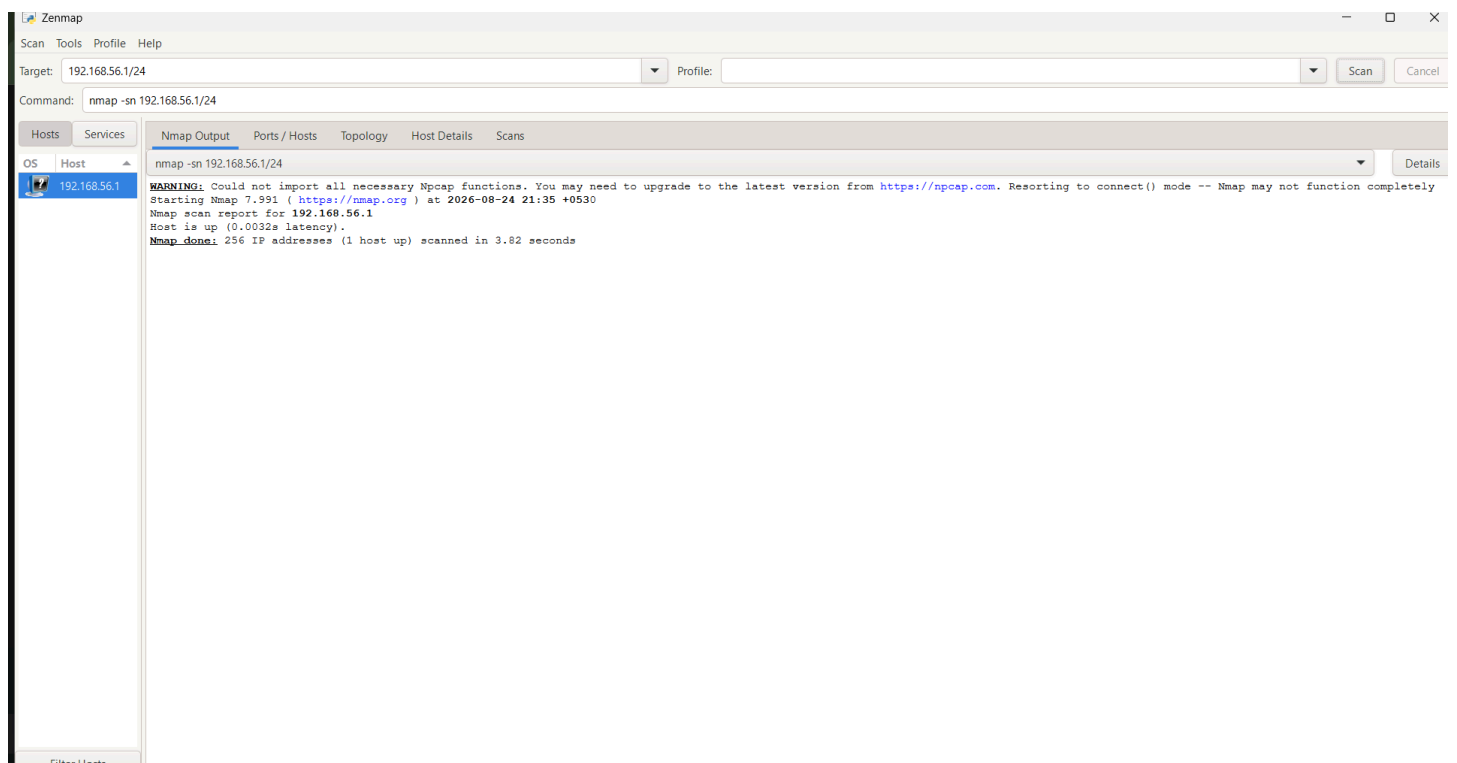


Toolchain / Environment: Zenmap GUI / Nmap (nmap -sn 192.168.56.1/24)

Technical Description & Observations:

- Shows execution of a ping/no-port scan (-sn) across the 256 addresses of the virtual adapter subnet.
- Highlights that exactly 1 host is active (192.168.56.1) with a low round-trip latency of 0.0032s.
- Notes the driver warning regarding Npcap functions defaulting to connect() mode fallback.

Figure 7: Local Area Network Ping Sweep & Active Node Discovery (10.153.99.0/24)

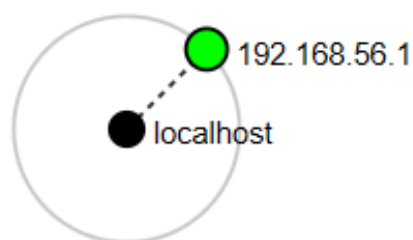


Toolchain / Environment: Zenmap GUI / Nmap (nmap -sn 10.153.99.24/24)

Technical Description & Observations:

- Demonstrates discovery sweeping on the local production subnet range.
- Confirms 2 active live hosts: the local host adapter node (10.153.99.24 at 0.00s latency) and an adjacent target node (10.153.99.99 at 0.0062s latency).

Figure 8: Single-Node Host-Only Topology Map (Subnet 192.168.56.0/24)

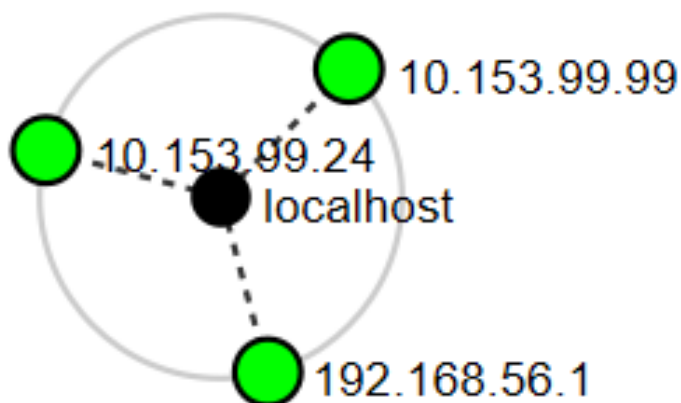


Toolchain / Environment: Zenmap Interactive Topology Viewer

Technical Description & Observations:

- Graphical network visualization showing the localhost root node connected directly to the virtual network adapter interface at IP 192.168.56.1.

Figure 9: Multi-Node Subnet Topology & Adjacent Node Map



Toolchain / Environment: Zenmap Interactive Topology Viewer

Technical Description & Observations:

- Multi-subnet radial topology chart linking the central localhost execution node across both broadcast segments:
 - The virtual host-only adapter endpoint: 192.168.56.1
 - The local network segment interface: 10.153.99.24
 - The active adjacent target node: 10.153.99.99

Conclusion for Screenshot & Artifact Analysis

The collected artifacts collectively illustrate an end-to-end reconnaissance and network discovery workflow, spanning external perimeter intelligence gathering down to internal broadcast domain mapping:

- 1. Comprehensive Surface Mapping (External vs. Internal):**
 - The passive OSINT and web fingerprinting phases (Figures 1 & 2) successfully identified external attack surfaces, exposed administrative metadata, routing ASNs, and public contact points across targeted domains (networkwalks.com and microsoft.com).
 - The local configuration and scanning phases (Figures 3, 6, 7, 8, & 9) validated internal interface parameters and verified active endpoints across distinct virtual and physical subnets (192.168.56.0/24 and 10.153.99.0/24).
- 2. Toolchain Integration & Operational Readiness:**
 - Deployment records (Figures 4 & 5) document the installation and dependency structure required for GUI-driven network scanning (Zenmap and Npcap).
 - The scanning outputs demonstrate how combining command-line utilities with graphical topology visualization aids in rapid asset discovery and network structure comprehension.

3. Key Security Takeaways:

- **Minimizing Information Leakage:** Publicly accessible technical metadata (e.g., exposed CMS versions, server tokens, and unsegmented subdomains) should be suppressed or migrated into private zones to hinder automated asset profiling.
- **Internal Segmentation:** Clear visibility into active subnet nodes reinforces the need for strict network segmentation and host-based access controls to prevent unauthorized lateral discovery.